

SMALLER LOOKS ROUNDER: RESOLUTION-DEPENDENT BIAS OF SHAPE DESCRIPTORS IN OPEN-SOURCE BIOIMAGE AND RADIOMICS SOFTWARE

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ABSTRACT

Circularity and sphericity are ubiquitous shape descriptors in cell morphometry and radiomics, yet their values depend on how the software estimates the perimeter or surface area of a mask. We audit eight implementations, including scikit-image (hence CellProfiler), OpenCV, ImageJ and PyRadiomics/IBSI, against exact ground truth and derive a closed-form model of their bias. The default 2D estimators make the same object rounder the fewer pixels it covers, and the IBSI sphericity of a sphere drops from 0.92 to 0.82 as CT slices thicken from 1 to 5 mm; isotropic resampling makes it worse. On nuclei re-segmented with Cellpose at several resolutions, halving the magnification shifts the default form factor by 0.7–0.95 standard deviations. In PanNuke, the default estimator nearly doubles the roundness difference between inflammatory and tumour nuclei, reverses the sign of another contrast at 10 $\times$, and inflates it 4.6-fold when scanners are mixed. A one-line correction of existing feature tables and a spacing-aware 3D Crofton estimator remove most of these errors.

Index Terms— Morphometry, radiomics, harmonization, digital geometry, reproducibility

1. INTRODUCTION

Shape descriptors of segmented objects are a staple of quantitative imaging. Nuclear circularity quantifies pleomorphism in histopathology, CellProfiler’s *FormFactor* is part of every Cell Painting profile [1, 2], and sphericity is a standardized radiomics feature of the Image Biomarker Standardisation Initiative (IBSI) [3]. Both are ratios of a size to a boundary measure, $FF = 4\pi A/P^2$ and $\Psi = (36\pi V^2)^{1/3}/S$, equal to 1 for a disk and a ball. Studies routinely pool data of different resolution: whole-slide images scanned at 20 $\times$ and 40 $\times$, cells of different sizes, CT series with 1–5 mm slices.

The area or volume of a mask is easy to estimate, since the pixel count is unbiased under random placement. The boundary is not: digital geometry has long shown that local perimeter estimators are biased and do not converge as the resolution increases [4, 5]. Kulpa [4] derived the large-object length factor κ of chain codes; Crofton estimators avoid it [6]; and MATLAB patched its circularity for small objects in 2023 [7].

What is missing is (a) a model of the *size dependence* of the estimators actually shipped in open-source tools, which is what breaks comparisons across resolutions, (b) evidence of how large the effect is in real pipelines, and (c) fixes that also repair data already analysed.

Contributions. (i) A closed-form model $P \approx \kappa(L - 2\pi\delta)$ with a new offset term $\delta = \sqrt{2}/\pi$ that predicts the size dependence of the default estimators (Fig. 1b). (ii) An audit of eight 2D/3D implementations against exact ground truth. (iii) Evidence on 66k annotated and 23k independently re-segmented nuclei and on lung tumours: resolution shifts, distorted biological contrasts, and degraded cross-resolution classification. (iv) Fixes: a post-hoc correction F1 that repairs existing feature tables without the masks; in 3D, we show that a spacing-aware Crofton estimator (F3) on the native grid removes the slice-thickness bias of IBSI sphericity, whereas IBSI’s recommended isotropic resampling does not.

2. BIAS MODEL AND FIXES

Let $X \subset \mathbb{R}^2$ have a smooth boundary of length L , digitized on a unit grid as $D = X \cap \mathbb{Z}^2$ (pixels whose centres lie in X) with random sub-pixel position and orientation. Most estimators trace a polygon along D whose edges take a few lattice directions (Fig. 1a). Let a boundary element of length ds and tangent angle θ be represented by polygon length $g(\theta) ds$, and let the polygon lie on average a distance δ inside ∂X . By Steiner’s formula, a closed curve moved inward by δ has length $L - 2\pi\delta$; averaging g over orientations then gives

$$P \approx \kappa(L - 2\pi\delta), \quad \kappa = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) d\theta. \quad (1)$$

Chain codes, used by scikit-image’s default `regionprops` perimeter (hence CellProfiler), by OpenCV’s `arcLength` and by MATLAB up to R2022b, join the centres of the boundary pixels (object pixels with a background 4-neighbour) by horizontal, vertical or diagonal steps of length 1 or $\sqrt{2}$. Thus $g(\theta) = |\cos \theta| + (\sqrt{2} - 1)|\sin \theta|$ for $|\theta| \leq \pi/4$ and $\kappa = 8(\sqrt{2} - 1)/\pi \approx 1.0548$ [4]. Along the grid axis closest to the boundary normal, the last pixel centre inside X lies at a uniform distance in $[0, 1)$ from ∂X , i.e. $\cos \theta/2$ on average perpendicular to it, so $\delta = \frac{4}{\pi} \int_0^{\pi/4} \frac{\cos \theta}{2} d\theta = \sqrt{2}/\pi \approx 0.45$.

For a disk of radius r pixels,

$$\text{FF}_{\text{chain}}(r) \approx \kappa^{-2} \left(r / (r - \sqrt{2}/\pi) \right)^2, \quad (2)$$

which exceeds 1 for $r < 8.7$ px and tends to 0.899: the same round nucleus measures 1.14 at $r = 4$ and 0.95 at $r = 16$ px. *Marching squares* (PyRadiomics 2D; IBSI meshes) place vertices on pixel-edge midpoints at the 0.5 level, giving the same κ but $\delta = 0$, i.e. a size-independent factor κ^{-2} on FF. *Pixel-edge polygons* (polygonized masks) follow the pixel borders: $g = |\cos \theta| + |\sin \theta|$, $\kappa = 4/\pi$, $\delta = 0$. ImageJ traces the pixel border but cuts staircase corners by diagonals, a hybrid. MATLAB R2023a corrects δ but not κ [7]. In 3D, marching cubes on a binary mask is the analogue of marching squares, but on anisotropic voxels its overestimation grows with the slice-to-pixel ratio (Fig. 1e). *Crofton* estimators count boundary crossings of lattice lines; by the Cauchy–Crofton formula (length or area from the mean number of crossings of random lines) they are unbiased apart from chords shorter than the sampling step [6].

F1: post-hoc correction. Inverting Eq. (1), $\hat{P} = P_{\text{chain}}/\kappa + 2\pi\delta'$ corrects any existing scikit-image/CellProfiler perimeter column without access to the masks. The first-order value $\delta' = \sqrt{2}/\pi$ is accurate for $r \geq 4$ px, but objects of 2–3 px lose extra length because the chain cuts strongly curved boundaries. We therefore calibrate δ' once on the synthetic shapes of Sec. 3 (minimax FF error over $r \in [2, 48]$ px), which gives 0.52, and use $\delta' = \frac{1}{2}$: $\hat{P} = P/1.0548 + \pi$. No real data are used for this choice. **F2:** for new 2D analyses, use the Crofton perimeter (available, but not default, in scikit-image).

F3: spacing-aware 3D Crofton estimator. For a mask with n voxels of volume v , consider the 13 lattice directions d (in physical units). Lines along d through the voxel centres cover the plane orthogonal to d with density $\|d\|/v$, so if N_d counts the object-to-background transitions along d , then $N_d v / \|d\|$ estimates the area of the object’s projection orthogonal to d (for non-convex objects, half the total crossing area). Cauchy’s formula, surface area = $4 \times$ the mean projected area over directions, gives

$$\hat{S} = 4 \sum_d w_d N_d v / \|d\|, \quad \hat{V} = n v, \quad (3)$$

where w_d is the fraction of the unit sphere closer to $\pm d/\|d\|$ than to any other direction, computed for the actual voxel spacing. This is the 13-direction estimator of [6] with spacing-dependent weights, as in MorphoLibJ [8]; it is not offered by PyRadiomics and needs no resampling.

3. EXPERIMENTS

Synthetic ground truth. We rasterize 5,000 shapes of known perimeter and area: ellipses (axis ratios 1, 0.6, 0.35) and two families of random smooth star-shaped “nuclei” (Fourier radius perturbations of orders 2–6), at equivalent radii of 2–48 px, with random rotation and sub-pixel offset. In 3D,

randomly rotated spheres and 1:0.75:0.5 ellipsoids (equivalent radius 3–20 mm) are digitized on $1 \times 1 \times s$ mm grids, $s \in [1, 5]$. We run the libraries’ own code (scikit-image 0.26 [9], OpenCV 4.13 [10], PyRadiomics 3.0.1 [11]); we verified in CellProfiler’s source that FormFactor uses scikit-image’s perimeter, ported ImageJ’s [12] traced perimeter line by line, and implemented MATLAB R2023a’s published formula [7].

Real nuclei. PanNuke fold 1 [13] (H&E, $40\times$, annotated nucleus types; 43,519 nuclei) and BBBC038 [14] (fluorescence and brightfield; 22,858 nuclei). Lower magnification is obtained in two ways. *Re-digitization*: each annotated mask is resampled on a $2\times$ or $4\times$ coarser grid (block mean ≥ 0.5), i.e. $20\times$ and $10\times$ for PanNuke. *Re-segmentation* (realistic, includes segmentation variability): the image is block-averaged and segmented independently at each resolution by the pre-trained Cellpose [15] *nuclei* model; nuclei are matched across resolutions by IoU ≥ 0.5 (1,500 PanNuke patches, 12,279 nuclei; 463 single-channel BBBC038 images, 10,454 nuclei). Nuclei touching the border, with holes, or fragmented are excluded. Since the true shape is unknown, we report the paired shift Δ/σ (mean change in FF over its native standard deviation; bootstrap s.e. ≈ 0.01) and Spearman’s ρ_A between area and FF. To assess biological conclusions, we compute Cohen’s d of FF between annotated PanNuke nucleus types, and a logistic regression predicting the type from area, solidity, eccentricity, extent, axis lengths and FF, trained at $40\times$ and tested at $10\times$ on held-out images (5 splits, balanced accuracy).

Lung tumours. The Medical Segmentation Decathlon lung set [16] (63 CTs) provides real tumour shapes. As its slice-wise annotations change when re-digitized, to isolate the estimator each tumour ($\geq 300 \text{ mm}^3$; 64 tumours, equivalent radius 3–45 mm) becomes a realistic phantom: the mask is resampled to 0.5 mm and smoothed ($\sigma = 1.5 \text{ mm}$), its 0.5 level set defines a surface whose true sphericity (0.60–0.96) is computed from a fine mesh, and this surface is point-sampled at 0.7 mm in-plane with 0.7–5 mm slices.

4. RESULTS

Synthetic audit (Fig. 1b). The chain-code tools overestimate circularity by +0.55 at $r = 2$ px and +0.11 at $r = 4$ px, cross zero near $r = 8$ and underestimate it by 0.08 at $r = 48$, as Eq. (2) predicts (dashed). ImageJ shows the same trend with a smaller range (+0.09 to -0.08). PyRadiomics and MATLAB R2023a are nearly size-independent for $r \geq 3$ px but offset by κ^{-2} (-0.08 to -0.09); pixel edges report 0.62 for a disk. Crofton and F1 are within ± 0.02 of the truth for $r \geq 4$ px.

Resolution shifts (Fig. 1c,d; Table 1). With the default estimator, halving the resolution shifts the mean FF by $0.7\text{--}0.95\sigma$ and quartering it by $2.5\text{--}5.6\sigma$, for re-digitized and re-segmented nuclei and both modalities; at $\frac{1}{4}$ resolution the median BBBC038 nucleus has an impossible

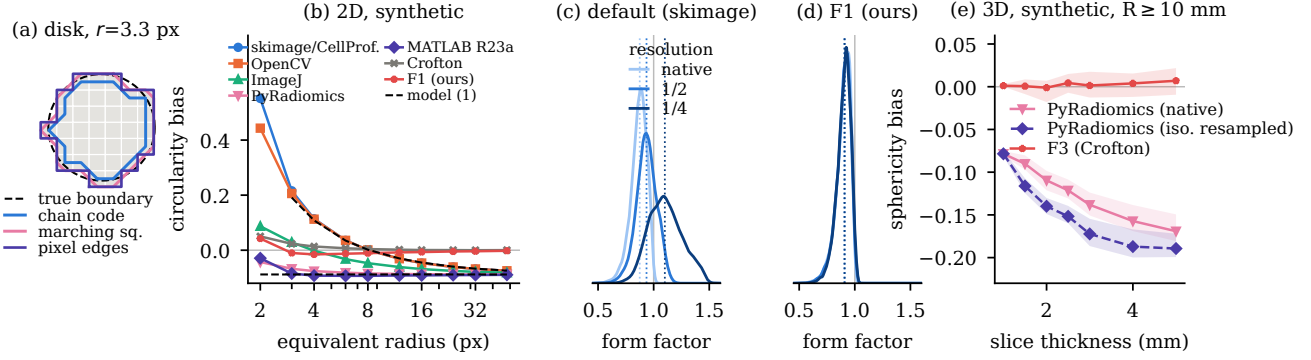


Fig. 1. (a) Polygons traced on a digitized disk: chain codes run inside the boundary ($\delta > 0$), marching squares on it, pixel edges outside. (b) Mean circularity bias on 5,000 synthetic shapes vs. size; dashed: model, Eq. (1). (c,d) FF of 10,454 BBBC038 nuclei segmented independently by Cellpose at native, $\frac{1}{2}$ and $\frac{1}{4}$ resolution (dotted: medians), with the default estimator (c) and F1 (d). (e) Sphericity bias of randomly rotated spheres/ellipsoids vs. slice thickness; bands: 10–90% range.

Table 1. Real nuclei: paired shift Δ/σ of FF when the resolution is halved ($\frac{1}{2}$) or quartered ($\frac{1}{4}$), by re-digitization of the annotations (“re-dig.”) or independent Cellpose re-segmentation (“re-seg.”), and area–FF correlation ρ_A at native resolution. Ideal: zero shift. OpenCV is within 0.1 of scikit-image except at $\frac{1}{4}$, PyRadiomics within 0.01 of Crofton.

FF from	PanNuke (H&E)					BBBC038 (fluorescence/brightfield)				
	re-dig.		re-seg.		ρ_A	re-dig.		re-seg.		ρ_A
	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$		$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	
scikit-image / CellProfiler	+0.72	+3.25	+0.68	+2.53	−0.34	+0.92	+5.56	+0.95	+3.30	−0.56
ImageJ	+0.34	+0.96	+0.31	+0.96	−0.21	+0.41	+1.31	+0.48	+1.68	−0.41
MATLAB R2023a	+0.06	+0.27	0.00	+0.16	−0.05	+0.11	+0.51	+0.04	+0.20	−0.19
Crofton	+0.11	+0.31	+0.06	+0.25	−0.09	+0.16	+0.51	+0.13	+0.46	−0.24
F1 (ours, post hoc)	+0.03	+0.16	−0.03	+0.05	−0.03	+0.07	+0.35	−0.01	+0.05	−0.16

Table 2. PanNuke: Cohen’s d of FF between nucleus types (inflammatory, connective vs. neoplastic; s.e. ≈ 0.014), within one magnification and when inflammatory nuclei at $20\times$ are compared with neoplastic ones at $40\times$ (“mixed”); balanced-accuracy drop of a $40\times$ -trained type classifier at $10\times$ (no FF: 0.008).

FF from	infl.-neo.		conn.-neo.		mixed	drop
	$40\times$	$10\times$	$40\times$	$10\times$		
scikit-image / CellProfiler	0.67	0.32	−0.62	+0.11	1.62	0.115
OpenCV	0.67	0.82	−0.62	−0.01	1.61	0.106
ImageJ	0.52	0.67	−0.71	−0.48	1.05	0.064
PyRadiomics, Crofton	0.40	0.53	−0.76	−0.62	0.56	0.041
F1 (ours)	0.35	0.39	−0.79	−0.41	0.35	0.054

FF > 1 . The default also creates an area–roundness correlation ($\rho_A = -0.34, -0.56$) that largely vanishes with F1 ($-0.03, -0.16$). Under re-segmentation, F1 reduces the shift to $|\Delta|/\sigma \leq 0.05$ at both resolutions, also below Crofton, whose missed short chords grow for small objects. Residual shifts under re-digitization are partly genuine, since coarse grids remove real boundary detail.

Biological contrasts (Table 2). Inflammatory nuclei are smaller than neoplastic ones (median radius 9.5 vs. 13.9 px at $40\times$), so the default estimator inflates their apparent extra roundness from $d = 0.35$ – 0.40 (size-stable estimators) to 0.67 . At $10\times$, the default reverses the sign of the connective–neoplastic contrast ($-0.62 \rightarrow +0.11$). In a study that mixes scanners (inflammatory nuclei at $20\times$, neoplastic at $40\times$), the default reports $d = 1.62$ instead of 0.35 (F1), a 4.6-fold inflation. A $40\times$ -trained type classifier loses 0.115 balanced accuracy at $10\times$ with the default FF (0.04 – 0.05 with size-stable estimators): a biased FF turns magnification into a domain shift.

3D (Fig. 1e). PyRadiomics sphericity is biased by -0.08 at 1 mm and -0.17 at 5 mm slices; isotropic resampling before extraction, as IBSI recommends, *increases* the bias to -0.19 . The mean bias of F3 stays ≤ 0.012 for $R \geq 10$ mm at every thickness (≤ 0.05 at $R = 5$ mm). On the 64 lung-tumour phantoms, PyRadiomics is biased by -0.065 at 0.7 mm and -0.136 at 5 mm slices, F3 by $+0.005$ and $+0.042$: a 3–13 $\times$ smaller error. F3’s residual stems from tumour poles thinner than a slice. Re-digitizing the raw annotations instead raises the mean sphericity at 5 mm for both estimators (PyRa-

diomics 0.57 $\rightarrow$ 0.66, F3 0.63 $\rightarrow$ 0.82), because thick slices remove genuine slice-wise contour jitter.

5. DISCUSSION AND RECOMMENDATIONS

The size dependence of the default circularity of scikit-image, CellProfiler and OpenCV is deterministic and predictable (Eq. (2)). It distorts any comparison of objects of different pixel size. Constant-offset estimators (PyRadiomics, MATLAB R2023a, pixel edges) mislead only absolute thresholds such as “circularity > 0.9”. In 3D, slice thickness, a known source of radiomic variability [17], acts on IBSI sphericity through the estimator itself. We recommend: (1) Crofton perimeters for new 2D analyses; (2) F1 to repair existing CellProfiler/scikit-image tables from their Area and Perimeter columns; (3) F3 for sphericity on anisotropic volumes, on the native grid; (4) reporting object sizes in pixels. Objects below ~ 3 px radius or two slices remain unreliable for any estimator, and block-averaging images only approximates re-imaging at lower magnification. Code: <https://github.com/Hannibal0319/isbi2027-shape-bias>.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study uses only public, anonymized datasets; no ethical approval was required.

7. ACKNOWLEDGMENTS

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